Summary in Graph

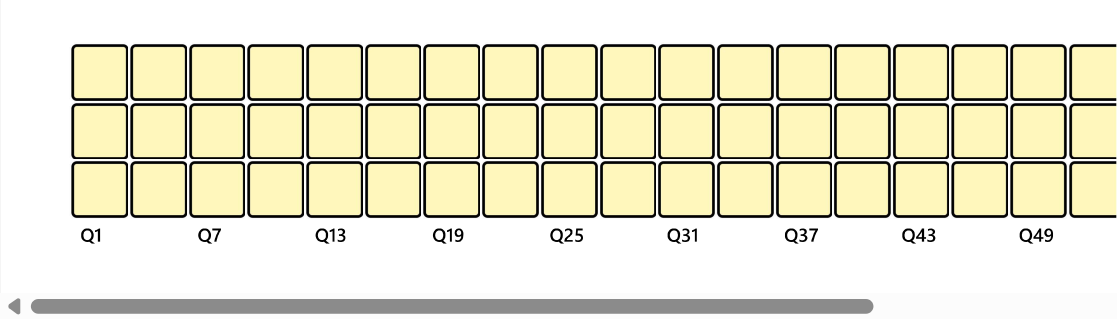
Exam Summary (GO Classes DA Test Series 2026 | MOCK GATE | Test 7)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	65 (30 + 35)
Total Marks:	100 (30 + 70)
Exam Duration:	180 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (15) out of 148 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Aptitude

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Verbal Aptitude

The village was nestled in a green spot, _____ the ocean and the hills.

A. through
B. in
C. at
D. between

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

If $\mathrm{md}(x) = |x|$,

- $\mathrm{mn}(x, y) =$ minimum of x and y, and
- $\mathrm{ma}(a, b, c) =$ maximum of a, b, c,

The value of the following expression:

$$\mathrm{ma}[\mathrm{md}(a), \mathrm{mn}(\mathrm{md}(b), a), \mathrm{mn}(ab, \mathrm{md}(ac))]$$

where $a = -2, b = -3$, and $c = 4$, is:

A. 2
B. 6
C. 8
D. -2

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

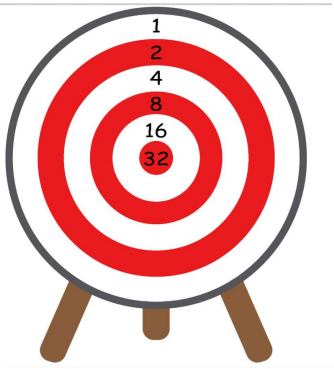
If $a - b = 3$ and $a^2 + b^2 = 29$, find the value of ab.

A. 10
B. 12
C. 15
D. 18

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Analytical Aptitude

Maryam is shooting arrows at the target shown. The number in a ring indicates how many points Maryam scores if an arrow hits that ring.



Maryam shoots three arrows and they all hit different rings.

Which total score is not possible for Maryam to obtain?

A. 56 points
B. 21 points
C. 10 points
D. 7 points

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

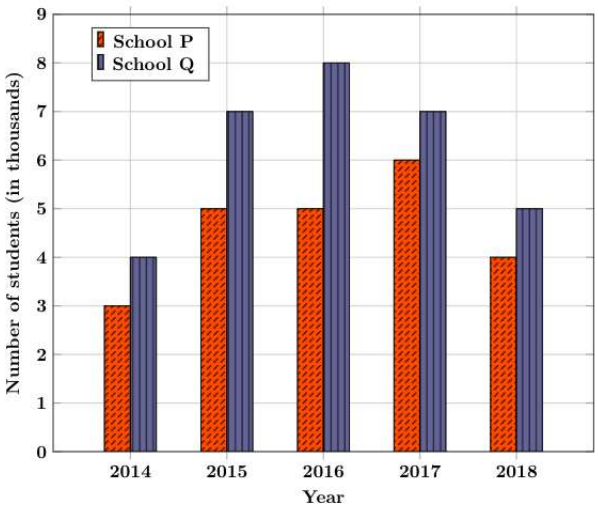
If a and b are positive numbers such that $a^b = b^a$ and $b = 9a$, then the value of a^4 is:

A. 9
B. $1/9$
C. 3
D. $\sqrt{6}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

The following figure shows the data of students enrolled in 5 years $(2014$ to $2018)$ for two schools P and Q. During this period, the ratio of the average number of the students enrolled in school P to the average of the difference of the number of students enrolled in schools P and Q is _____.



A. $8 : 23$
B. $23 : 8$
C. $23 : 31$
D. $31 : 23$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 05sec | Discuss

Q #7 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

The salaries A, B, C are in the ratio $2 : 3 : 5$. If the increments of $15\%, 10\%$ and 20% are allowed respectively in their salaries, then what will be new ratio of their salaries?

A. $3 : 3 : 10$

B. $10 : 11 : 20$
C. $23 : 33 : 60$
D. Cannot be determined

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

For any natural numbers m, n, and k, such that k divides both $m + 2n$ and $3m + 4n\mathrm{k}$ must be a common divisor of

A. m and n
B. $2m$ and $3n$
C. $2m$ and n
D. m and $2n$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

If x and y are positive real numbers such that $\log_x\left(x^2 + 12\right) = 4$ and $3\log_y x = 1$, then $x + y$ equals

A. 11
B. 20
C. 10
D. 68

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

A trader has 50 kg of rice, a part of which he sells at 14% profit and the rest at 6% loss. On the whole his loss is 4%. What is the quantity sold at 14% profit and that at 6% loss?

A. $2, 48$
B. $4, 46$
C. $5, 45$
D. $60, 20$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let the joint probability mass function of discrete random variables X and Y be given by:

$$p(x, y) = \frac{1}{25}\left(x^2 + y^2\right) \quad \text{where} \quad x = 1, 2 \quad \text{and} \quad y = 0, 1, 2$$

Find $P\{X > Y\}$.

A. $\frac{1}{5}$

B. $\frac{2}{5}$

C. $\frac{3}{5}$

D. $\frac{4}{5}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let $\Phi(x)$ be the cumulative distribution function (CDF) of a standard normal random variable (mean 0, variance 1), which gives the probability that a standard normal random variable is less than x. Which of the following expressions are TRUE?

A. $\Phi(-x) = \Phi(x)$
B. $-\Phi(x) + \Phi(-x) = 1$
C. $\Phi(-x) = \frac{1}{\Phi(x)}$
D. $\Phi(x) + \Phi(-x) = 1$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let $\mathbb{X}$ be a continuous random variable with cumulative distribution function (CDF) $F_{\mathbb{X}}(u)$. Which of the following statements are true?

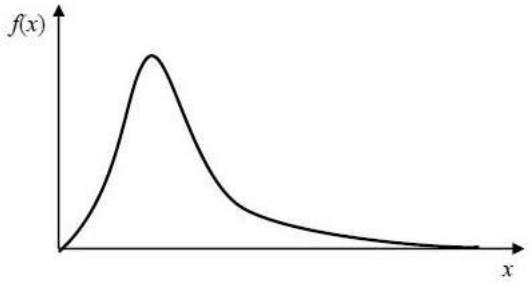
I. If $b > a$, then $F_{\mathbb{X}}(b) > F_{\mathbb{X}}(a)$
II. If $F_{\mathbb{X}}(b) > F_{\mathbb{X}}(a)$, then $b > a$
III. $F_{\mathbb{X}}(\alpha) = \frac{1}{2}$ for some $\alpha, -\infty < \alpha < \infty$

A. III only
B. I and III only
C. II and III only
D. I, II, and III

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

A probability distribution with right skew is shown in the figure.



The correct statement for the probability distribution is

A. Mean is equal to mode
B. Mean is greater than median but less than mode
C. Mean is greater than median and mode
D. Mode is greater than median

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Numerical Type | Award: 1 | Penalty: 0 | Linear Algebra

Consider the following matrix with 10 rows:

$$\begin{bmatrix} \longleftarrow & \text{row } 1 & \longrightarrow \\ \longleftarrow & \text{row } 2 & \longrightarrow \\ \vdots & \vdots & \vdots \\ \longleftarrow & \text{row } 10 & \longrightarrow \end{bmatrix}$$

This matrix represents 10 rows: row 1, row $2, \ldots$, row 10. What is the minimum number of elementary row exchanges required to put the rows in reverse order (i.e., row 10, row $9, \ldots$, row 1)?

Your Answer: | Correct Answer: 5 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Which of the following statements regarding matrix operations and vector spaces is/are FALSE?

A. If A and B are 3×3 matrices and $B = (\mathbf{b}_1 \mathbf{b}_2 \mathbf{b}_3)$, then $AB = (A\mathbf{b}_1 + A\mathbf{b}_2 + A\mathbf{b}_3)$.
B. The set of all vectors of the form $\begin{pmatrix} 3a+b \\ 4 \\ a-5b \end{pmatrix}$, where a and b are arbitrary real numbers, is a vector space.
C. $\mathbb{R}^2$ is a subspace of $\mathbb{R}^3$.
D. If U is the echelon form of A, then $\text{Col}\, U = \text{Col}\, A$, where Col denotes the column space.

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Consider two parallel hyperplanes in R^d :

$$H_1 : \mathbf{w}^T\mathbf{x} = +3,$$
$$H_2 : \mathbf{w}^T\mathbf{x} = -2,$$

What is the distance between H_1 and H_2 ?

A. 5
B. $5/\|\mathbf{w}\|$
C. $5/\|\mathbf{w}\|^2$
D. $5 \cdot \|\mathbf{w}\|$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

If $f(x) = 4e^{3x}$, compute $\lim_{h \to 0} \dfrac{f(2+h) - f(2)}{h}$.

A. 0
B. $12e^6$
C. $6e^6$
D. $4e^6$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

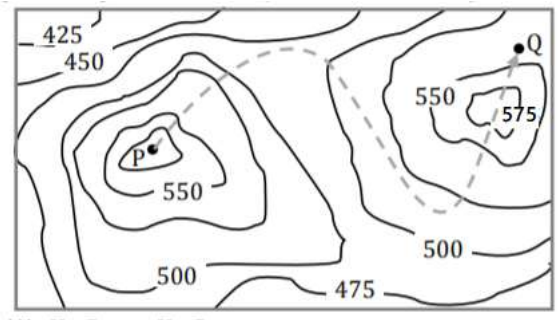
If $f(4) = 10$ and $f'(x) \geq 3$ for $2 \leq x \leq 4$, then the Mean Value Theorem guarantees $f(2)$ can be no bigger than

A. 1
B. 2
C. 4
D. 8

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

A contour line joins locations having the same height above the mean sea level. The following is a contour plot of a geographical region. Contour lines are shown at 25 m intervals in this plot.



The path from P to Q is best described by

A. Up-Down-Up-Down
B. Down-Up-Down-Up
C. Down-Up-Down

D. Up-Down-Up

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Consider the following 2-dimensional data, where $\hat{y} = \sigma\left(x^T\theta\right)$ represents the predicted probability of each example belonging to class 1. The true label for each example is given by y, where $y = 1$ indicates the example belongs to class 1, and $y = 0$ indicates it belongs to class 0. The data and predicted probabilities are as follows:

x_1	x_2	y	$\hat{y} = \sigma(x^T\theta)$
0.35	−0.75	0	0.56
−0.95	1.20	0	0.31
1.10	−1.45	1	0.87

What is the mean cross-entropy loss on just the above three rows of our training data?

A. $-\frac{1}{3}(\log(0.56) + \log(0.31) + \log(0.87))$

B. $-\frac{1}{3}(\log(0.44) + \log(0.69) + \log(0.87))$

C. $-\frac{1}{3}(\log(0.56) + \log(0.69) + \log(0.87))$

D. $-\frac{1}{3}(\log(0.44) + \log(0.31) + \log(0.87))$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Select Type | Award: 1 | Penalty: 0 | Machine Learning

Consider the standard two-class SVM with the hinge loss. For a given value of the regularization parameter C, the following inequality holds for the leave-one-out (LOO) error:

$$\text{LOO error } \leq T$$

where T is a value that depends on the number of support vectors (\#SVs) and the number of training examples l.

Which of the following is/are the correct expression(s) for T ?

A. $T = \frac{\#\text{SVs}}{l}$

B. $T = \frac{\#\text{SVs}}{l} + 1$

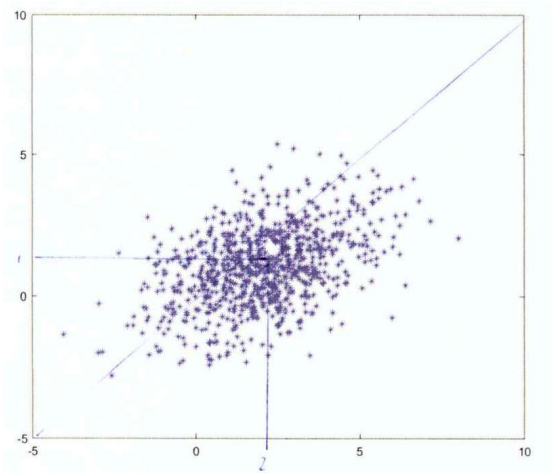
C. $T = \frac{\#\text{SVs}}{l} - 1$

D. $T = \#\text{SVs}$

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Consider the following dataset.



Which covariance matrix best describes the covariance in this dataset?

A. $\begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$

B. $\begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix}$

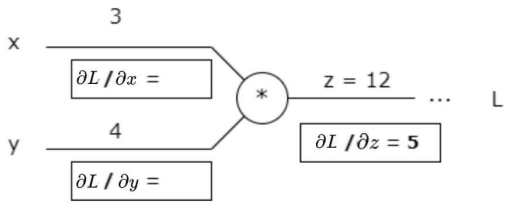
C. $\begin{bmatrix} 0 & 2 \\ 3 & 0 \end{bmatrix}$

D. $\begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Below is a simple computation graph with inputs x and y where an initial computation of $z = x \cdot y$ occurs before the unknown path to the final loss L.



A forward propagation pass has been completed with values $x = 3$ and $y = 4$, and the upstream gradient is given as $\frac{\partial L}{\partial z} = 5$.

Complete the backpropagation pass by filling in the scalar answers to boxes $\frac{\partial L}{\partial x}$ and $\frac{\partial L}{\partial y}$.

A. $\frac{\partial L}{\partial x} = 20, \quad \frac{\partial L}{\partial y} = 15$

B. $\frac{\partial L}{\partial x} = 15, \quad \frac{\partial L}{\partial y} = 20$

C. $\frac{\partial L}{\partial x} = 48, \quad \frac{\partial L}{\partial y} = 36$

D. $\frac{\partial L}{\partial x} = 36, \quad \frac{\partial L}{\partial y} = 48$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Given 6 data points in a 5-dimensional space, we can represent these data points by a 6×5 matrix X, where each row corresponds to a data point. The Singular Value Decomposition (SVD) for this matrix is given by:

$$X = \begin{bmatrix} \longleftarrow & \text{training data 1} & \longrightarrow \\ \longleftarrow & \text{training data 2} & \longrightarrow \\ \vdots & \vdots & \vdots \\ \longleftarrow & \text{training data 6} & \longrightarrow \end{bmatrix} = \begin{bmatrix} a & 0 \\ -3a & 0 \\ 2a & 0 \\ 0 & b \\ 0 & -2b \\ 0 & b \end{bmatrix} \times \begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{bmatrix} \times \begin{bmatrix} c & c & c & 0 & 0 \\ 0 & 0 & 0 & d & d \end{bmatrix}$$

Where $\sigma_1 > \sigma_2$. What is the direction of the first principal component for the original data points?

A. $\pm[a, -3a, 2a, 0, 0]$
B. $\pm[c, c, c, 0, 0]$
C. $\pm[a, b, c, 0, 0]$
D. $\pm[0, 0, 0, c, c]$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Consider two statements $\mathrm{S1}$ and $\mathrm{S2}$ below.

- $\mathrm{S1}$: An $\Theta(n \log n)$ algorithm ALWAYS runs faster than an $\Theta\left(n^2\right)$ algorithm.
- $\mathrm{S2}$: An $\Theta(n \log n)$ algorithm ALWAYS runs slower than an $\Theta\left(n^2\right)$ algorithm.

Which of the following is correct ?

A. $\mathrm{S1}$ is true but $\mathrm{S2}$ is false
B. $\mathrm{S1}$ is false but $\mathrm{S2}$ is true
C. Both statements are true
D. Both Statements are false

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Which is the FASTEST algorithm for finding the shortest path between two vertices in an UNWEIGHTED graph?

A. Dijkstra's shortest path
B. Depth-first search
C. Breadth-first search
D. Bellman-Ford

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

Which of the following statements is true about the differences between Equal-Width and Equal-Frequency Binning?

A. Equal-Width Binning ensures each bin has the same number of data points, while Equal-Frequency Binning divides the range equally.
B. Equal-Frequency Binning divides the data into bins with the same number of data points, while Equal-Width Binning divides the range equally.
C. Equal-Width Binning and Equal-Frequency Binning both ensure each bin has the same range of values.
D. Equal-Width Binning guarantees bins with the same number of data points, whereas Equal-Frequency Binning divides the data evenly by range.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Numerical Type | Award: 1 | Penalty: 0 | Programming in Python

Consider the following Python code:

```
x = 22

def reset_x():
    x = 0

reset_x()
print(x)
```

What will be the output of the code when executed?

Your Answer: | Correct Answer: 22 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Artificial Intelligence

Consider the game of Pacman in the absence of ghosts and with a single food dot in the maze. In state s, Pacman is located at position (i_s, j_s) while the food dot is located at (x_s, y_s). At timestep t, the score of the game is $500 - t$. The game ends when the food is eaten.

Which of the following heuristics is not admissible?

A. $h(s) = |i_s - x_s| + |j_s - y_s|$
B. $h(s) = \sqrt{(i_s - x_s)^2 + (j_s - y_s)^2}$
C. $h(s) = (i_s - x_s)^2 + (j_s - y_s)^2$
D. $h(s) = \min(|i_s - x_s|, |j_s - y_s|)$

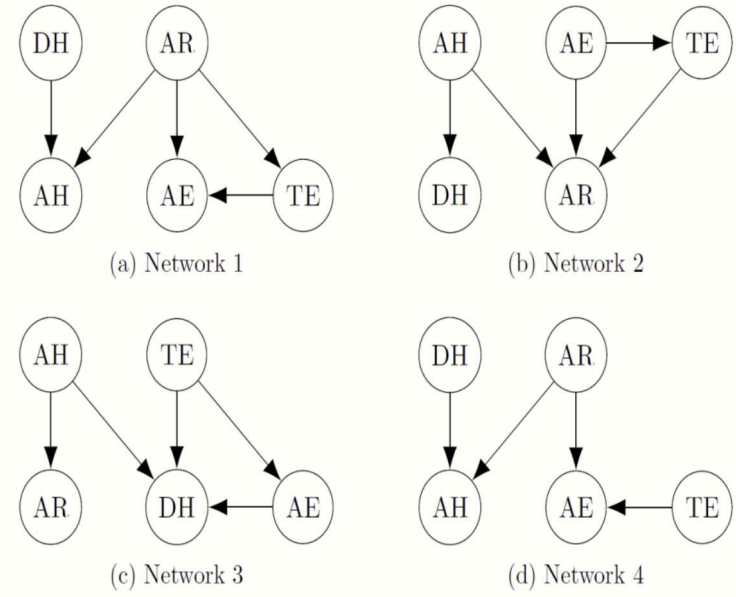
Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Artificial Intelligence

Consider the following problem example:

Aching elbows and aching hands may be the result of arthritis. Arthritis is also a possible cause of tennis elbow, which in turn may cause aching elbows. Dishpan hands may also cause aching hands. Let AR stand for "arthritis", AH for "aching hands", AE for "aching elbow", TE for "tennis elbow", and DH for "dishpan hands".

Which of the following Bayesian networks represent the causal links described in the problem example defined above?



A. Network 1
B. Network 2
C. Network 3
D. Network 4

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 | Multiple Select Type | Award: 1 | Penalty: 0 | Artificial Intelligence

In the context of adversarial search, Alpha-Beta $(\alpha - \beta)$ pruning is an optimization technique applied to the Minimax algorithm to reduce the number of nodes that need to be evaluated.

Which of the following statements about Alpha-Beta pruning is/are true?

A. $\alpha - \beta$ pruning is generally faster than minimax, but loses the guarantee of optimality.
B. $\alpha - \beta$ pruning can reduce computation time by pruning portions of the game tree.
C. $\alpha - \beta$ pruning always returns the same value as minimax for all nodes on the leftmost edge of the tree, assuming successor game states are expanded from left to right.
D. $\alpha - \beta$ pruning always returns the same value as minimax for all nodes of the tree.

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Artificial Intelligence

A knowledge base KB contains the following statement (i.e. considered to be true).

$$\neg(W \wedge R) \Rightarrow (Q \vee T)$$

The question is whether we can prove $Q \vee T$ from this KB. Which of the following answers is correct?

A. Yes, we can prove $Q \vee T$.
B. No, we cannot derive $Q \vee T$, but if we add the premiss W to KB the statement $Q \vee T$ can be derived.
C. No, we cannot derive $Q \vee T$, but if we add the premiss $\neg W$ to KB the statement $Q \vee T$ can be derived.
D. None of the above.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

Consider the set F of following functional dependencies on relation $R(ABCDE)$:

$$\begin{aligned} AB &\rightarrow C \\ BC &\rightarrow E \\ BD &\rightarrow E \\ C &\rightarrow B \\ D &\rightarrow A \end{aligned}$$

The relation R is in

A. Not in $2NF$
B. $2NF$ but not $3NF$
C. $3NF$ but not BCNF
D. BCNF

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Numerical Type | Award: 1 | Penalty: 0 | Databases

Consider an empty table REPRESENTATIVE whose schema is

```
(pid, name, dept, rank, age).
```

The table has two candidate keys: { name } and { dept, rank }. Now, assume that we try to insert the following tuples in the top-down order. If an insertion is rejected by the database (for violation of the candidate key constraint), we ignore the tuple and move on to the next one. Then, eventually, what will be the content of the table?

- (1, Adam, CS, asso, 43)
- (2, Adam, EE, asst, 37)
- (3, Bob, EE, full, 56)
- (4, Calvin, CS, asso, 45)
- (5, Emily, EE, asst, 28)

At the end, the number of tuples in the table is ?

Your Answer: | Correct Answer: 3 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

A wallet contains two biased coins, with probabilities of heads being p_1 and p_2, respectively. One coin is chosen at random, and the other one is discarded. The chosen coin is tossed once and comes up heads. What is the probability that it will come up heads again on the second toss, given that it came up heads the first time?

A. $\frac{p_1^2 + p_2^2}{p_1 + p_2}$

B. $\frac{p_1 + p_2}{p_1^2 + p_2^2}$

C. $\frac{p_1^2}{p_1 + p_2}$

D. $\frac{p_1 + p_2^2}{p_1^2 + p_2}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Consider a uniform probability distribution over the interval $[0, 1]$, where the sample space is $S = [0, 1]$. Let A and B be two events, where:

- $A = \left[0, \frac{1}{2}\right]$ is the event where the outcome is in the first half of the interval.
- $B = \left[\frac{1}{4}, \frac{5}{8}\right]$ is the event where the outcome is between $\frac{1}{4}$ and $\frac{5}{8}$.

What is the value of $P[B \mid A]$, the probability of B occurring given that A has occurred?

A. $\frac{1}{2}$

B. $\frac{3}{8}$

C. 1

D. $\frac{5}{8}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Multiple Select Type | Award: 2 | Penalty: 0 | Probability

Consider the joint probability mass function of the random variables X, Y, and Z given by:

$$f(1, 2, 3) = f(2, 1, 1) = f(2, 2, 1) = f(2, 3, 2) = \frac{1}{4}$$

and $f(i, j, k) = 0$ for all other (i, j, k).

Which of the following statements about the marginal probability mass functions of X, Y, and Z is/are true?

A. $P(X = 1) = \frac{1}{4}, P(X = 2) = \frac{3}{4}$

B. $P(Y = 1) = \frac{1}{4}, P(Y = 2) = \frac{1}{2}, P(Y = 3) = \frac{1}{4}$

C. $P(Z=1)=\frac{1}{2}, P(Z=2)=\frac{1}{4}, P(Z=3)=\frac{1}{4}$

D. $P(X=1)=\frac{1}{4}, P(Y=2)=\frac{1}{2}, P(Z=3)=\frac{1}{4}$

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Select Type | Award: 2 | Penalty: 0 | Probability

Let $\mathbb{X}$ be a continuous random variable with pdf $f_{\mathbb{X}}(u)$ satisfying:

$$f_{\mathbb{X}}(u) = f_{\mathbb{X}}(-u) \text{ for all } u, -\infty < u < \infty$$

Suppose that $\text{var}(\mathbb{X}) = 9$. Let $\mathbb{Y} = |\mathbb{X}|$ and $\mathbb{Z} = -\mathbb{X}$. Which of the following statements is/are true?

A. $\mathbb{E}\left[\mathbb{Y}^2\right] = 9$
B. $\mathbb{E}[\mathbb{Y}] = 3$
C. $\mathbb{E}\left[\mathbb{Z}^2\right] = 9$
D. $\mathbb{E}[\mathbb{Z}] = 0$

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

A program selects a random integer x as follows:

- A random bit $(0 \text{ or } 1)$ is first generated uniformly, with equal probabilities.
- If the bit is 0, x is drawn uniformly from the set $\{0, 1, 2, 3\}$.
- If the bit is 1, x is drawn uniformly from the set $\{0, -1, -2, -3, -4, -5, -6, -7\}$.

If we get an x from the program with $|x| = 1$, what is the probability that x is negative?

A. $\frac{1}{3}$

B. $\frac{1}{4}$

C. $\frac{1}{2}$

D. $\frac{1}{12}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #31 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

If $A_1, A_2, \ldots, A_n$ are independent events with probabilities $p_1, p_2, \ldots, p_n$ respectively, then $P(\cup_{i=1}^{n} A_i)$ equals

A. $\Sigma_{i=1}^{n}\ p_i$
B. $\Pi_{i=1}^{n}\ p_i$
C. $\Pi_{i=1}^{n}\ (1-p_i)$
D. $1-\Pi_{i=1}^{n}\ (1-p_i)$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #32 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Let $\boldsymbol{A}$ be a 3×4 matrix and $\boldsymbol{B}$ be a 4×3 matrix with real entries such that $\boldsymbol{AB}$ is non-singular. Consider the following statements:

- P: Nullity of $\boldsymbol{A}$ is $\mathbf{0}$.
- Q: $\boldsymbol{BA}$ is a non-singular matrix.

Then

A. Both P and Q are TRUE
B. P is TRUE and Q is FALSE
C. P is FALSE and Q is TRUE
D. Both P and Q are FALSE

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #33 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Consider the following statements:

- S_1: If $\mathbf{x}$ is not in a subspace W, then $\mathbf{x} - \mathrm{proj}_W \mathbf{x} = 0$.
- S_2: A least-squares solution of $A\mathbf{x} = \mathbf{b}$ is a vector $\hat{\mathbf{x}}$ such that $\|\mathbf{b} - A\mathbf{x}\| \geq \|\mathbf{b} - A\hat{\mathbf{x}}\|$ for all $\mathbf{x}$ in $\mathbb{R}^n$.

Which one of the following choices is correct?

A. Both S_1 and S_2 are true.
B. S_1 is true,S_2 is false.
C. S_1 is false, S_2 is true.
D. Both S_1 and S_2 are false.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #34 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Given the vectors:

$$\mathbf{v}_1 = \begin{pmatrix} 4 \\ 1 \\ 3 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 2 \\ -2 \\ -1 \end{pmatrix}, \quad \mathbf{w}_1 = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}, \quad \mathbf{w}_2 = \begin{pmatrix} -6 \\ 2 \\ -1 \end{pmatrix}$$

Let:

$$A = (\mathbf{v}_1 \mathbf{v}_2), \quad B = (\mathbf{w}_1 \mathbf{w}_2)$$

Which of the following statement(s) is TRUE?

A. The column spaces of A and B are identical, so $\mathbf{v}_1, \mathbf{v}_2$ and $\mathbf{w}_1, \mathbf{w}_2$ are two bases for the same subspace V.
B. The column spaces of A and B are not identical, so $\mathbf{v}_1, \mathbf{v}_2$ and $\mathbf{w}_1, \mathbf{w}_2$ are not two bases for the same subspace.
C. The set $\{\mathbf{v}_1, \mathbf{v}_2\}$ is a basis for $\mathbb{R}^2$.
D. The set $\{\mathbf{w}_1, \mathbf{w}_2\}$ is linearly dependent, so it cannot be a basis for any subspace.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #35 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Let R be the row reduced echelon form of a 4×4 real matrix A and let the third column of R be $\begin{bmatrix} 0 \\ 1 \\ 0 \\ \mathbf{0} \end{bmatrix}$.

Consider the following statements:

- P: If $\begin{bmatrix} \alpha \\ \beta \\ \gamma \\ 0 \end{bmatrix}$ is a solution of $\boldsymbol{Ax} = \mathbf{0}$, then $\boldsymbol{\gamma} = \mathbf{0}$.
- Q: For all $b \in \mathbb{R}^4$, $\text{rank}[A \mid b] = \text{rank}[R \mid b]$

Then

A. Both P and Q are TRUE
B. P is TRUE and Q is FALSE
C. P is FALSE and Q is TRUE
D. Both P and Q are FALSE

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #36 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be a linear transformation satisfying

$$T(1,0,0) = (0,1,1), T(1,1,0) = (1,0,1), T(1,1,1) = (1,1,2)$$

Then

A. T is one-one but T is NOT onto
B. T is one-one and onto
C. T is NEITHER one-one NOR onto
D. T is NOT one-one but T is onto

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #37 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Let A and B be matrices. Two matrices B and A are said to be similar if $B = S^{-1}AS$ for some matrix S. Consider the following two statements:

- S1: A singular matrix A cannot be similar to a non-singular matrix B.
- S2: AB and BA are similar matrices.

Which of the following is true?

A. $S1$ is true but $S2$ is false.
B. $S1$ is false but $S2$ is true.
C. Both $S1$ and $S2$ are true.
D. Both $S1$ and $S2$ are false.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #38 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Let λ be a constant, $x \in \mathbb{R}^2$ and define function

$$f(x) = x_1^2 - 4x_1x_2 + 3x_2^2 + \lambda\|x\|_2^2.$$

What is the minimum λ to make $f(x)$ a convex function?

A. $2 - \sqrt{5}$
B. $2 + \sqrt{5}$
C. $-2 + \sqrt{5}$
D. $-2 - \sqrt{5}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #39 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Consider the following loss function on vectors $w \in \mathbb{R}^4$:

$$L(w) = w_1^2 + 2w_2^2 + w_3^2 - 2w_3w_4 + w_4^2 + 2w_1 - 4w_2 + 4$$

What are the values of w that minimize $L(w)$, and is the solution unique?

A. $(-1, 1, 0, 0)$, and the solution is unique.
B. $(-1, 1, 1, 1)$, and the solution is unique.
C. Any point of the form $(-1, 1, x, x)$, and the solution is not unique.
D. Any point of the form $(1, -1, x, x)$, and the solution is not unique.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #40 | Multiple Select Type | Award: 2 | Penalty: 0 | Machine Learning

Consider the mistake bound of the perceptron algorithm. Which of the following statements are correct? Select ALL below that are always true.

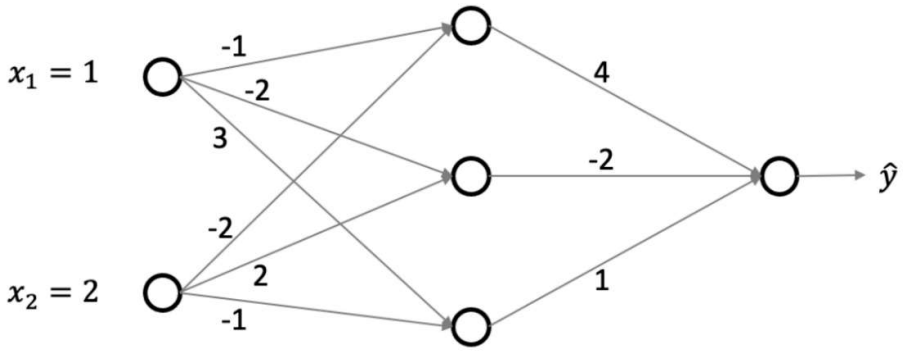
A. If the minimum distance from any data point to the separating hyperplane is increased, without any other change to the data points, the mistake bound will also increase.
B. If the whole dataset is shifted away from the origin, then the mistake bound will also increase.
C. If the pair-wise distance between data points is increased, i.e., the data is scaled by some constant value, then the mistake bound will also increase.
D. The mistake bound is linearly inverse-proportional to the minimum distance of any data point to the separating hyperplane of the data.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #41 | Numerical Type | Award: 2 | Penalty: 0 | Machine Learning

Consider the following neural network with weights shown in the image below. Every hidden neuron uses the ReLU activation function, and there is no activation function on the output neuron. Assume there are no bias terms.

What is the output of this network with the input $x = (1, 2)$?



Your Answer: | Correct Answer: -3 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #42 | Numerical Type | Award: 2 | Penalty: 0 | Machine Learning

We ran the perceptron learning algorithm with offset (bias) on the following dataset. The initial value of the bias θ_0 is assumed to be 0. For each data point, the number of mistakes made by the algorithm is recorded.

Without actually running the algorithm, determine the final value of θ_0.

i	1	2	3	4	5
$x^{(i)}$	(-4, 3)	(-2, 2)	(-2, 0)	(1, 3)	(2, 0)
$y^{(i)}$	1	-1	-1	-1	-1
# mistakes	2	3	2	1	0

Your Answer: | Correct Answer: -4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #43 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Consider the following problem, where we have two classes: $\{\text{Tainted}, \text{Clean}\}$, and three covariate features: (a_1, a_2, a_3). These attributes are also binary variables: $a_1 \in \{$ on, off $\}, a_2 \in \{\text{blue}, \text{red}\}, a_3 \in \{$ light, heavy $\}$.

We are given a training set as follows:

- Tainted: (on, blue, light), (off, red, light), (on, red, heavy)
- Clean: (off, red, heavy), (off, blue, light), (on, blue, heavy)

There are three examples with class "Tainted" and three examples with class "Clean."

Now, consider a new example $\mathbf{x} =$(on, red, light). What is $p($ class $=$ Tainted $|\mathbf{x})$?

A. $\frac{8}{9}$

B. $\frac{1}{2}$

C. $\frac{3}{4}$

D. $\frac{1}{4}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #44 | Numerical Type | Award: 2 | Penalty: 0 | Programming in Python

Consider the following Python code that generates all subsets of a given list:

```
def generate_subsets(lst):
    if not lst:
        return [[]]
    subsets = generate_subsets(lst[1:])
    return subsets + [[lst[0]] + subset for subset in subsets]

my_list = [1, 2, 3]
subsets = generate_subsets(my_list)
print(len(subsets))
```

What will be the output of the code when executed?

Your Answer: | Correct Answer: 8 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #45 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Suppose you are given an array A with $2n$ numbers.

The numbers in odd positions are sorted in ascending order, that is, $A[1] \leq A[3] \leq \ldots \leq A[2n-1]$.

The numbers in even positions are sorted in descending order, that is, $A[2] \geq A[4] \geq \ldots \geq A[2n]$.

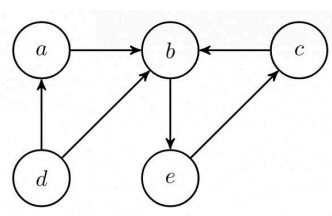
What is the method you would recommend for determining if a given number is in the array?

A. Sort the array using quick-sort and then use binary search.
B. Merge the sorted lists and perform binary search.
C. Perform a single binary search on the entire array.
D. Perform separate binary searches on the odd positions and the even positions.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #46 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Consider the following directed graph.



Suppose a depth-first traversal of this graph is performed, assuming that whenever there is a choice, the vertex earlier in the alphabetical order is to be chosen. Suppose the number of tree edges is T, the number of back edges is B and the number of cross edges is C. Then

A. $B = 1$, $C = 1$, and $T = 4$.
B. $B = 0$, $C = 2$, and $T = 4$.
C. $B = 2$, $C = 1$, and $T = 3$.
D. $B = 1$, $C = 2$, and $T = 3$.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #47 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Insertion sort performs best case when input is sorted and worst case when input is sorted backwards. Compute the average COST of insertion sort to sort a vector with n different integers when with probability $\frac{\log n}{n}$ one chooses a vector which is sorted backwards and with probability $1 - \frac{\log n}{n}$ one chooses a sorted vector.

A. $\Theta\left(n^3\right)$
B. $\Theta(n \log n)$
C. $\Theta(n)$
D. $\Theta\left(n^2\right)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #48 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

A stable sort preserves the order of values that are equal with respect to the comparison function. We have a list of three-dimensional points

$[(7, 1, 8), (3, 5, 7), (6, 1, 4), (6, 5, 9), (0, 2, 5), (9, 0, 9)]$.

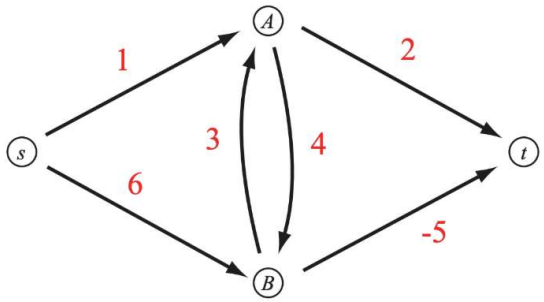
We sort these in ascending order by the second coordinate. Which of the following corresponds to a stable sort of this input?

A. $[(9, 0, 9), (7, 1, 8), (6, 1, 4), (0, 2, 5), (6, 5, 9), (3, 5, 7)]$
B. $[(0, 2, 5), (3, 5, 7), (6, 1, 4), (6, 5, 9), (7, 1, 8), (9, 0, 9)]$
C. $[(9, 0, 9), (7, 1, 8), (6, 1, 4), (0, 2, 5), (3, 5, 7), (6, 5, 9)]$
D. $[(9, 0, 9), (6, 1, 4), (7, 1, 8), (0, 2, 5), (3, 5, 7), (6, 5, 9)]$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #49 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Consider the following weighted graph below:



Which of the following options is/are correct about the given graph?

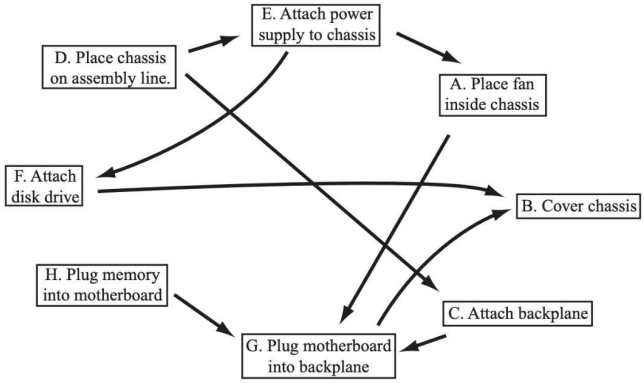
Assume s is the source node for all the options.

A. Dijkstra's algorithm will find the correct shortest path to all the vertices.
B. Dijkstra's algorithm will NOT be able to find the correct shortest path to one or more vertices.
C. Bellman-Ford algorithm will find the correct shortest path to all the vertices.
D. Bellman-Ford algorithm will NOT be able to find the correct shortest path to one or more vertices.

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #50 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

In the following graph, boxes represent tasks that must be performed in the assembly of a computer, and arrows represent constraints that one task must be performed before another (for instance, the disk drive must be attached before the chassis can be covered).



If there are t tasks and c constraints (edges), what is the time complexity for solving this problem using the best possible algorithm?

A. $\Theta(t + c)$
B. $\Theta(t \log c)$
C. $\Theta(c \log t)$
D. $\Theta(tc)$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #51 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

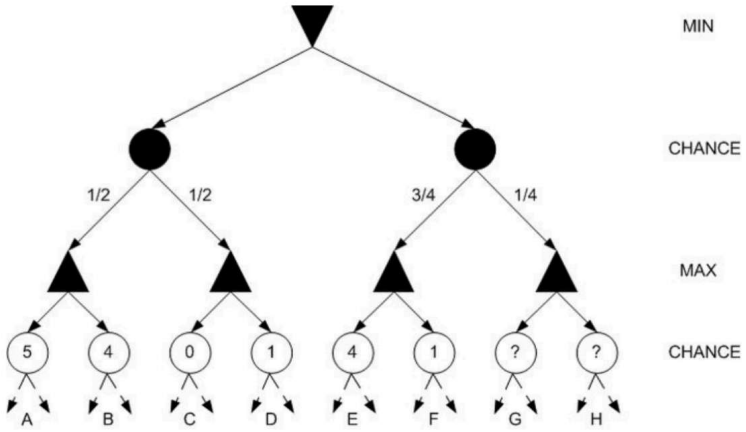
Consider a one-sided test regarding some population parameter μ. Suppose the hypotheses are $H_0 : \mu \leq \mu_0$ vs $H_1 : \mu > \mu_0$, the test statistic is called T, and the null hypothesis is rejected whenever $T \geq c$. Suppose the true value of the parameter is $\mu = \mu_1$. Which expression correctly expresses the power of the test (check all that apply)?

A. $1 - P(T \leq c \mid \mu = \mu_0)$
B. $P(T \leq c \mid \mu = \mu_0)$
C. $1 - P(T \leq c \mid \mu = \mu_1)$
D. $P(T \leq c \mid \mu = \mu_1)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #52 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Artificial Intelligence

Consider the following game tree with an element of chance.



The letters under the bottom row of chance nodes are labels for the nodes just above the letter.

One can also apply (alpha-beta) pruning on this game tree. The numbers inside the chance nodes on the bottom row are the computed values of these chance nodes.

Which of the following statements is correct?

A. If the value of node G is larger or equal than 0 then pruning will never calculate the value of node H.
B. If the value of node G is smaller than 0 then pruning will always calculate the value of node H.
C. If the value of node G is larger or equal than 0 then pruning will always always calculate the value of node H.
D. None of the above.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #53 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Artificial Intelligence

In the context of the Valentine puzzle the predicate $\mathrm{Send}(x, c, y)$ means "person x sends card c to person y ".

What is implied by the following formula in First Order Predicate Logic?

$$\forall x_1, x_2, c_1, c_2, y \ \mathrm{Send}(x_1, c_1, y) \wedge \mathrm{Send}(x_2, c_2, y) \Rightarrow x_1 = x_2$$

A. Every person gets a card from at most one person.
B. Every person sends a card to at most one person.
C. Every person sends at most one card.
D. Every card is sent by at most one person.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #54 | Numerical Type | Award: 2 | Penalty: 0 | Databases

Consider the following table whose name is T :

A	B	C
1	10	100
2	10	10
3	40	100
4	30	200
5	25	90

The number of rows in the output of the following SQL query is ?

```
SELECT A
FROM T AS R
WHERE NOT EXISTS (
    SELECT *
    FROM T
    WHERE T.B >= R.B AND T.C >= R.C
);
```

Your Answer: | Correct Answer: 0 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #55 | Numerical Type | Award: 2 | Penalty: 0 | Databases

B^{+} Tree is constructed by parameter n

- Each Node (except root) has $\lceil n/2 \rceil$ to n pointers
- Each Node (except root) has $\lceil n/2 \rceil - 1$ to $n - 1$ search-key values

What is the total number of key values in the internal nodes of B^{+} tree with 1000 Leaf Nodes?

Your Answer: | Correct Answer: 999 | Not Attempted | Time taken: 00min 00sec | Discuss

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